V2I DIGITAL TWIN

Telemetry Overhead

in Vehicular Edge Computing Networks

Technical report | B.Tech 5th semester assignment

Abstract

This study quantifies the channel-time and bandwidth overhead of continuously synchronizing vehicle state to roadside digital twins. A one-road SUMO scenario supplies mobility samples for five tested densities and three upload intervals. A separate log-distance V2I link model estimates theoretical Shannon capacity, and a deterministic shared-channel FCFS queue converts that capacity into packet service times and deadline outcomes. The saved experiment contains 30 configurations, 120 system and RSU summary rows, and 1,110 per-vehicle rows. In those saved results, packet delivery ratio is 1.0 throughout; maximum system-wide utilization is 5.1092%, while maximum individual-RSU utilization is 10.1023%. No system or RSU case exceeds the strict 60% threshold over the tested 5 to 100 vehicle range. These results describe the stated idealized model, not measured radio-network performance.

Problem

Frequent Digital Twin updates consume uplink airtime that would otherwise be available for edge-task offloading. The overhead depends on update frequency, vehicle-to-RSU link quality, and the number of vehicles sharing each RSU channel.

Objectives

Measure modeled telemetry utilization, packet delivery ratio, residual bandwidth, and per-vehicle upload rate; compare density, update interval, and path-loss exponent; and identify whether any tested case exceeds 60% utilization.

Scenario and experimental design

SUMO mobility foundation and the evaluated configuration matrix

Road and RSUs

One straight, single-lane, 1,000 m road. Three RSUs are fixed at x = 250 m, 500 m, and 750 m. Vehicles are assigned to their nearest RSU using one-dimensional along-road distance.

SUMO observation

The saved configuration specifies a 0.05 s simulation step and a 120 s observation duration. Vehicles start from rest at evenly spaced positions across the first 75% of the road; the same mobility trace is reused across upload intervals and exponents.

Factor matrix

Five densities x three upload intervals x two path-loss exponents give 30 configurations. n = 3.0 is the 15-case core matrix; n = 3.5 is the 15-case sensitivity run.

Vehicles	Intervals	Exponents	Configurations
5, 10, 20, 50, 100	50, 100, 500 ms	3.0, 3.5	30 total
Summary rows	Per-vehicle rows	RSU channels	Observation
120 (30 system + 90 RSU)	1,110	3 x 10 MHz	120 s; 50 ms step

The summary CSV reports each configuration at system scope and once per RSU. The per-vehicle CSV stores attempted packets, observed active time, and transmissions per second.

Source configuration: config/telemetry_experiments.json. Road and RSU positions come from config/smoke_test.json.

V2I channel model

Unit-explicit link budget and theoretical capacity

Parameter	Configured value
Carrier frequency	5.9 GHz
Channel bandwidth	10 MHz = 10,000,000 Hz
Vehicle transmit power	20 dBm
Noise PSD	-174 dBm/Hz
Receiver noise figure	9 dB
Reference distance	1 m
Reference path loss	Friis free-space at 1 m = 47.8648 dB
Path-loss exponent	n = 3.0 or 3.5
Telemetry payload	32 bytes = 256 bits

Wavelength lambda = c / f; c = 299,792,458 m/s and f = 5.9 x 10^9 Hz.

Reference loss PL(d0) = 20 log10(4 pi d0 / lambda) = 47.8648 dB for d0 = 1 m.

Log-distance path loss PL(d) = PL(d0) + 10 n log10(d / d0), for d >= d0.

Received power Pr (dBm) = Pt (dBm) - PL(d) (dB).

Receiver noise N (dBm) = N0 (dBm/Hz) + 10 log10(B in Hz) + NF (dB) = -95 dBm.

SNR SNRdB = Pr - N; SNRlinear = 10^(SNRdB / 10).

Shannon capacity C = B log2(1 + SNRlinear), in bits/s. This is theoretical link capacity, not measured throughput.

Payload service time t = 32 x 8 / C seconds, an ideal payload serialization time used by the queue model.

The one-dimensional model uses absolute x-distance to the nearest RSU. Integrated simulation distances below 1 m are clamped to the reference distance. The channel calculation omits fading, shadowing, antenna gains, interference, collisions, PHY errors, and retransmissions.

Telemetry packets and shared-channel access

How periodic state updates become packet outcomes

Field	Bytes	Contents represented
Position	12	Vehicle location
Speed	4	Vehicle speed
Heading	4	Travel direction
Acceleration	4	Longitudinal acceleration
Timestamp	8	State sample time
Total	32	256 payload bits

1. Each RSU has one independent, shared 10 MHz uplink channel.

2. A perfectly coordinated, non-preemptive FCFS queue serves one packet at a time per RSU. Independent RSUs can serve packets concurrently.

3. Packets with equal generation times are ordered by vehicle ID, then packet ID. Arrival concurrency therefore creates a queue, not a collision.

4. The service duration is 256 payload bits divided by that vehicle-to-RSU Shannon capacity. No protocol headers or MAC overhead are added.

5. A packet is delivered only if serialization completes by its next-upload deadline (generation time + interval) and by the 120 s observation end. A packet missing either boundary is dropped before transmission and consumes no airtime.

6. Queue deadline misses and observation-end misses are the only modeled packet-loss causes. There are no collision, interference, random PHY-error, or retransmission processes.

This deterministic queue is a capacity/deadline abstraction, not a full physical-layer or standards-compliant MAC simulation.

Metrics and saved results

CSV-derived summary at the highest tested density

Metric	Definition and scope
RSU utilization (%)	100 x delivered serialization seconds at that RSU / 120 s.
System utilization (%)	100 x sum of delivered airtime over 3 RSUs / (3 x 120 s); the average RSU busy fraction.
PDR	Delivered / attempted packets, by RSU or system. CSV range: 1.000 to 1.000.
RSU residual (MHz)	10 x (1 - RSU utilization / 100).
System residual (MHz)	Sum of the three RSU residuals, at most 30 MHz.
Individual rate (1/s)	One vehicle: attempts / active seconds. CSV: transmissions_per_second.
Weighted system rate (1/s)	Total attempts / total active vehicle-seconds. CSV: mean_transmissions_per_vehicle_second.

Maximum system utilization: 5.1092% | Maximum individual RSU: 10.1023% (rsu1). Both at 100 vehicles, 50 ms, n=3.5.

System-wide results at 100 vehicles

Interval (ms)	n	Util. (%)	PDR	Residual (MHz)	Weighted system rate (1/s)
50	3.0	0.6236	1.000	29.8129	19.996
50	3.5	5.1092	1.000	28.4673	19.996
100	3.0	0.3119	1.000	29.9064	10.002
100	3.5	2.5555	1.000	29.2333	10.002
500	3.0	0.0625	1.000	29.9812	2.003
500	3.5	0.5126	1.000	29.8462	2.003

Utilization

Saved plot from results/; plotted values are drawn from the saved summary CSV.

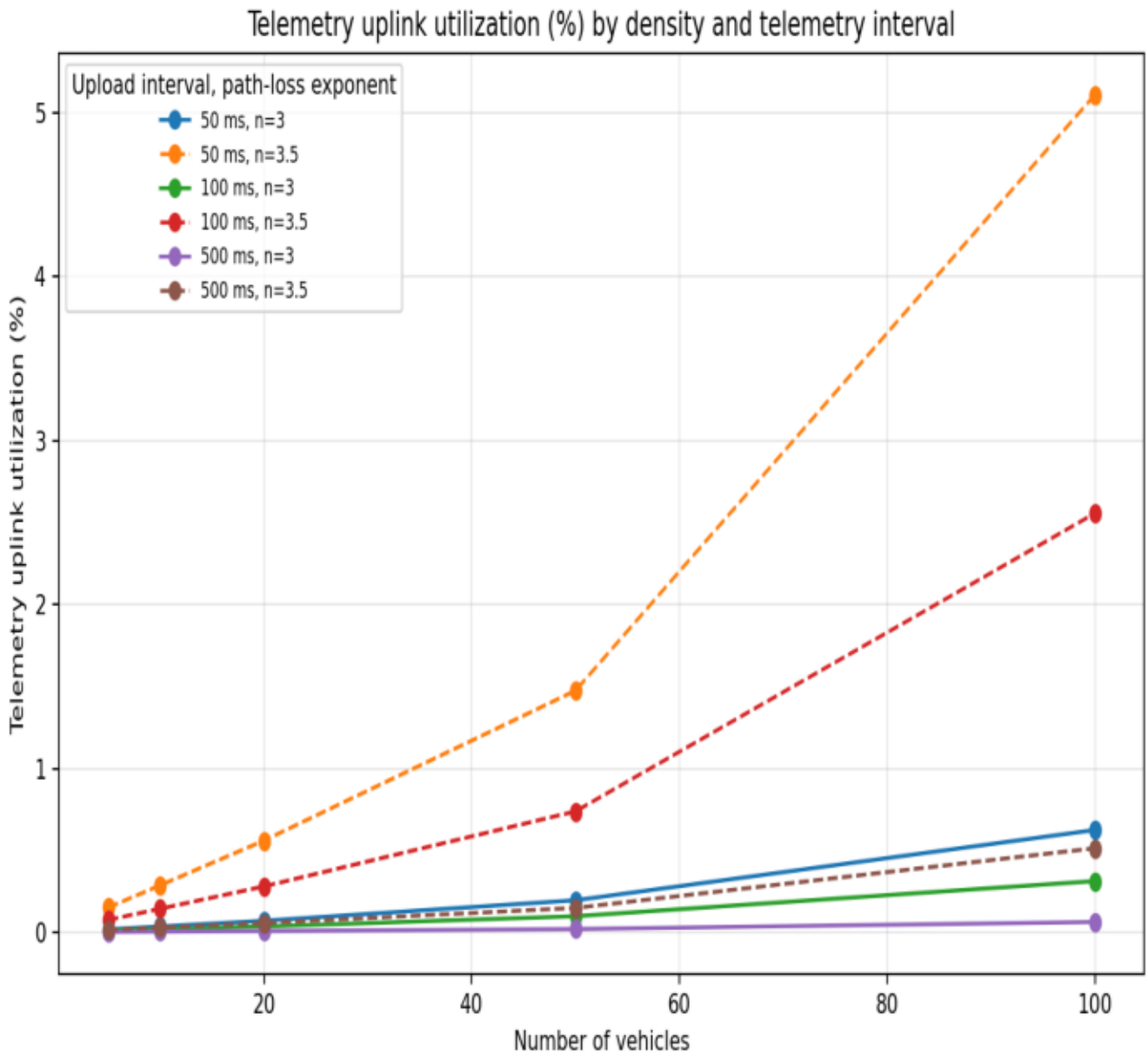


Figure 1. Utilization by vehicle density, upload interval, and path-loss exponent.

System-wide telemetry utilization is the average busy fraction across the three independent 10 MHz RSU channels.

Packet delivery ratio

Saved plot from results/; plotted values are drawn from the saved summary CSV.

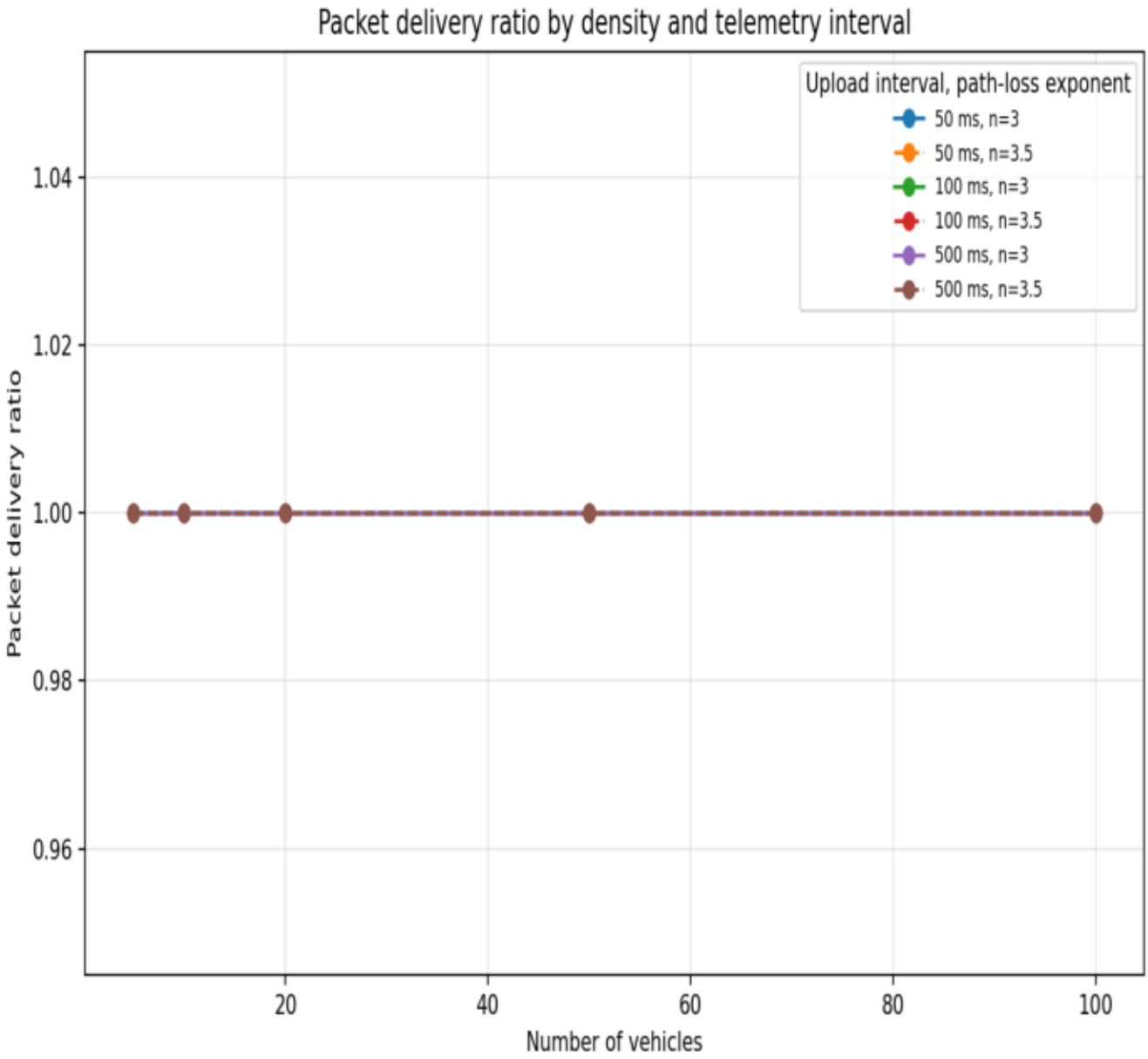


Figure 2. Packet delivery ratio by vehicle density, upload interval, and path-loss exponent.

PDR is delivered packets divided by attempted packets. The saved matrix reports 1.0 in every configuration under the stated deadline queue model.

Residual bandwidth

Saved plot from results/; plotted values are drawn from the saved summary CSV.

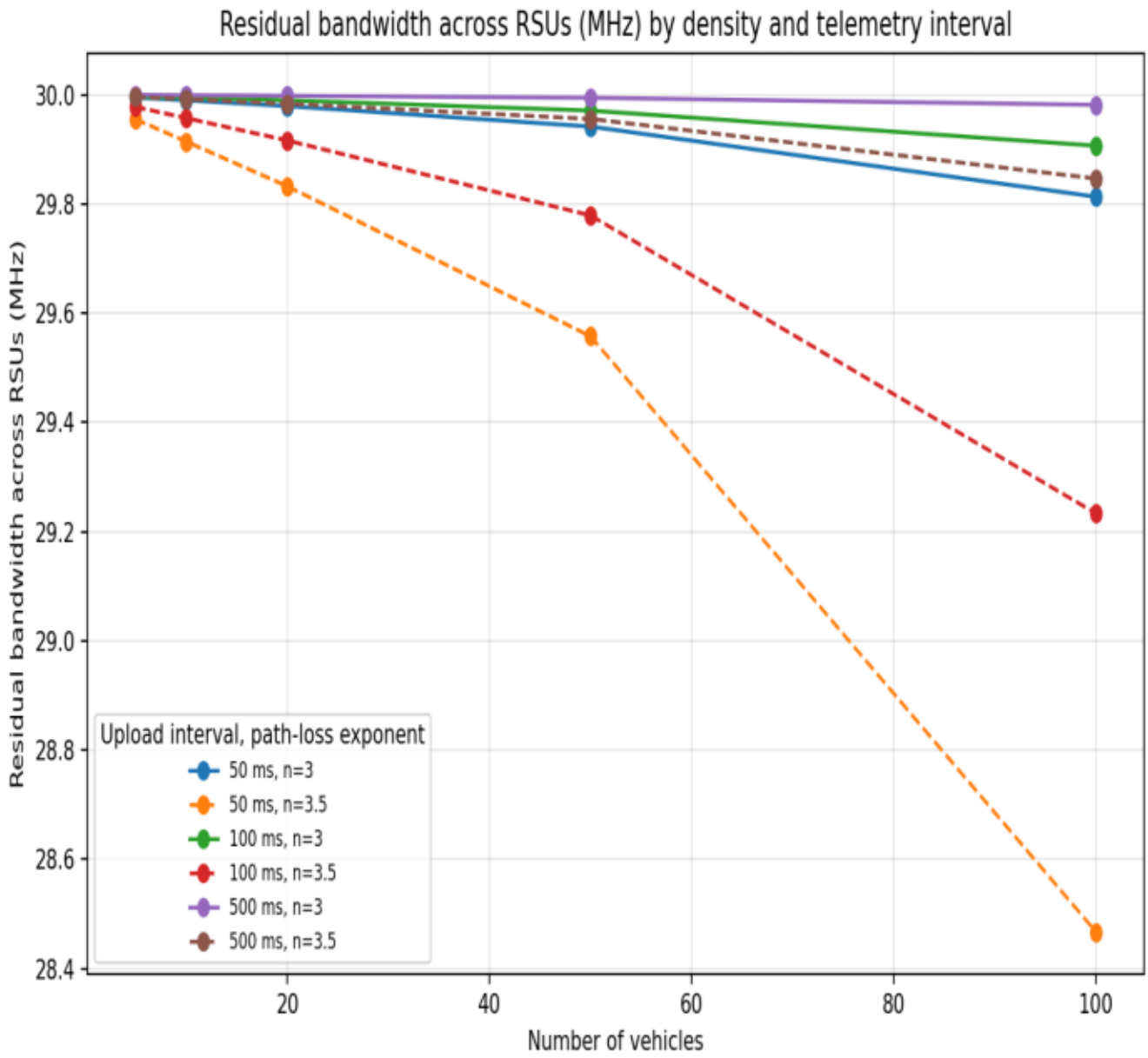


Figure 3. Residual bandwidth by vehicle density, upload interval, and path-loss exponent.

System residual bandwidth is the sum of the three RSU residuals, so its nominal maximum is 30 MHz.

Weighted system transmission rate

Saved plot from results/; plotted values are drawn from the saved summary CSV.

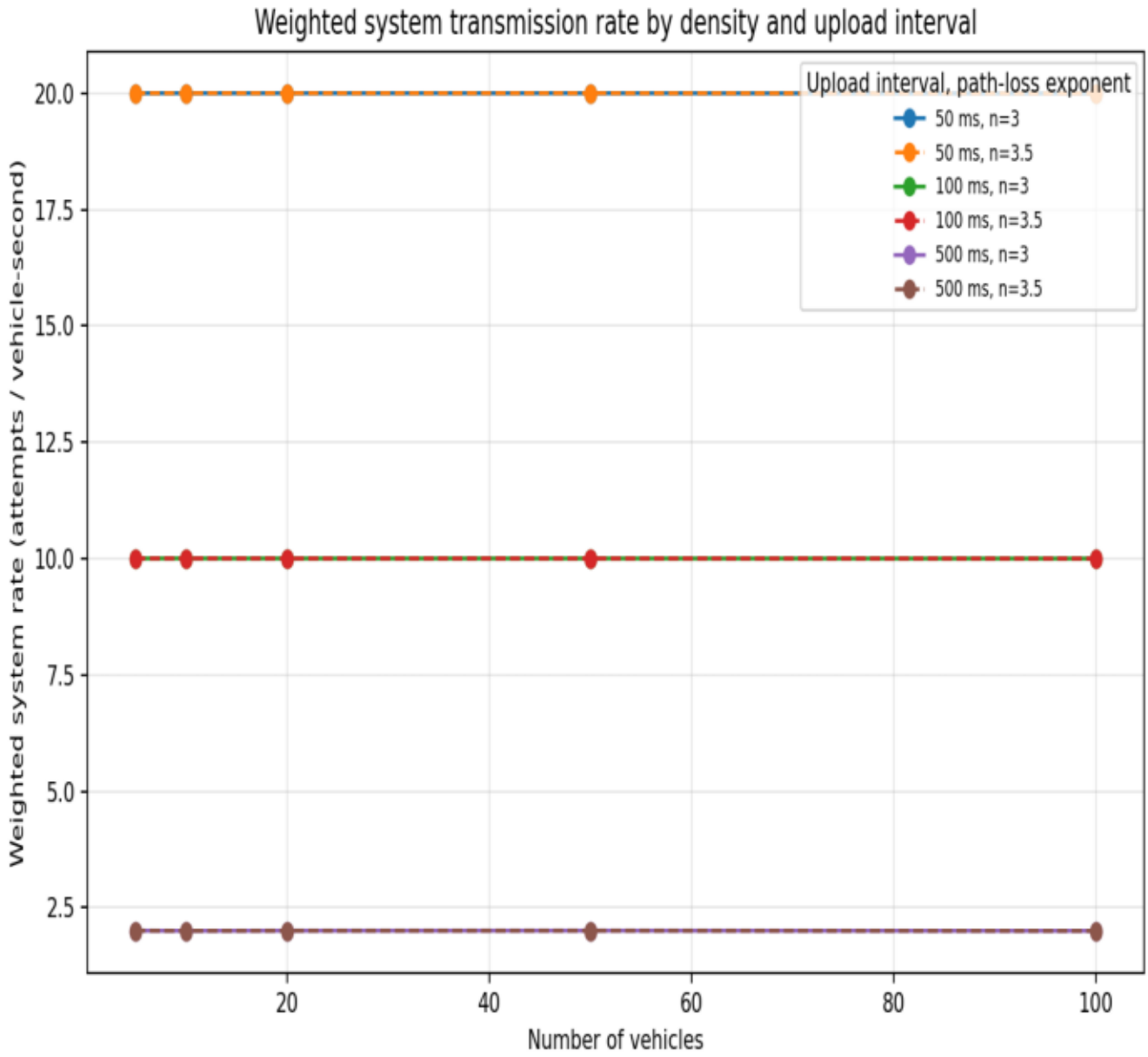


Figure 4. Weighted system rate by density, upload interval, and path-loss exponent.

This plot shows total attempted packets divided by total active vehicle-seconds. It is an active-time-weighted system aggregate, not an unweighted arithmetic mean or a series of individual vehicle rates.

Saturation threshold and interpretation

Strict 60% comparison, evaluated separately by interval, exponent, and scope

Interval (ms)	n	Scope	First >60%	Max observed
50	3	System-wide	No crossing	0.6236%
50	3	rsu1	No crossing	1.1150%
50	3	rsu2	No crossing	0.2442%
50	3	rsu3	No crossing	0.5117%
50	3.5	System-wide	No crossing	5.1092%
50	3.5	rsu1	No crossing	10.1023%
50	3.5	rsu2	No crossing	0.8785%
50	3.5	rsu3	No crossing	4.3466%
100	3	System-wide	No crossing	0.3119%
100	3	rsu1	No crossing	0.5577%
100	3	rsu2	No crossing	0.1222%
100	3	rsu3	No crossing	0.2560%
100	3.5	System-wide	No crossing	2.5555%
100	3.5	rsu1	No crossing	5.0528%
100	3.5	rsu2	No crossing	0.4395%
100	3.5	rsu3	No crossing	2.1743%
500	3	System-wide	No crossing	0.0625%
500	3	rsu1	No crossing	0.1118%
500	3	rsu2	No crossing	0.0245%
500	3	rsu3	No crossing	0.0513%
500	3.5	System-wide	No crossing	0.5126%
500	3.5	rsu1	No crossing	1.0130%
500	3.5	rsu2	No crossing	0.0881%
500	3.5	rsu3	No crossing	0.4369%

Threshold definition: for each interval, exponent, and scope, select the first tested vehicle count with utilization strictly greater than 60.0%; equality does not qualify. No saved value equals or exceeds 60.0%. All 24 scope-specific groups return no threshold over 5, 10, 20, 50, and 100 vehicles.

Telemetry overhead rises with more frequent uploads and larger path-loss exponent, but remains below the threshold in this tested scenario. These results do not establish a general network saturation density.

Limitations and conclusion

Results are model outputs, not field measurements

Model limitations

The link model is one-dimensional and line-of-sight, with large-scale log-distance loss and no shadowing, fading, antenna gains, interference, or blockage. Shannon capacity is an ideal upper bound used to form payload service time; it is not observed throughput. The access model assumes perfect centralized FCFS scheduling and independent RSU channels. It has no collisions, random PHY errors, retransmissions, protocol headers, or realistic MAC control behavior. Consequently, PDR = 1.0 means that every packet met the modeled deadline in the saved scenario; it is not a real-world delivery guarantee. Residual bandwidth is an idle-time share translated to MHz and does not simulate task-offload traffic or its own scheduling.

Conclusion

For the saved 30-case experiment, utilization remains below 60% at both scopes for every tested density from 5 through 100 vehicles. The observed maxima are 5.1092% system-wide and 10.1023% for RSU rsu1; both occur at 100 vehicles, 50 ms uploads, and n=3.5. PDR remains 1.0 across the saved summary. No saturation threshold was observed within the tested matrix. The matrix and plots were already present in the workspace; creating this report did not rerun SUMO.

Per-vehicle transmission rates in the saved CSV

Upload interval	Nominal rate	Observed CSV range
50 ms	20 /s	19.9917 to 20.0000 /s
100 ms	10 /s	10.0000 to 10.0233 /s
500 ms	2 /s	2.0000 to 2.0390 /s

Ranges include all 1,110 vehicle rows. The denominator is each vehicle's observed active time, so finite trace boundaries produce small differences from nominal rates.

Reproduction commands and files

PowerShell commands from the project root

Run and validation commands

Full SUMO experiment (rewrites results)

```
.\.venv\Scripts\python.exe src\telemetry_experiment.py
```

Plots only from saved summary CSV

```
.\.venv\Scripts\python.exe src\telemetry_experiment.py --plots-only
```

Full pytest suite

```
.\.venv\Scripts\python.exe -m pytest -q
```

SUMO + TraCI smoke test

```
.\.venv\Scripts\python.exe src\sumo_smoke_test.py
```

Channel example at n = 3.0

```
.\.venv\Scripts\python.exe src\channel_simulation.py --path-loss-exponent 3.0
```

```
Channel at n = 3.5: .\.venv\Scripts\python.exe src\channel_simulation.py --path-loss-exponent 3.5
```

Saved inputs: results/telemetry_summary.csv, results/telemetry_per_vehicle.csv, and the four results/*.png plots. Experiment parameters are in config/telemetry_experiments.json.